

About Hud



Hud (Seidu) Daannaa

Manager - Sr Security Architect (R&D) - Lead (Big Data & SOC Specialist) - Lecturer - MSc| CEH| AWS-SAA| AZ-500| GCHQ| EC-Council-CS-Career-Mentor

With over a **decade** of experience in cybersecurity, **Hud specializes** in building cutting-edge security tools and secure architectures. Recently, He developed a **state-of-the-art threat intelligence SaaS platform leveraging AI technologies**. As an experienced Security Architect, He led and **managed the development of advanced Next Generation SOC's for the private and government sectors**, all with **top-level security clearance**, ensuring robust defence mechanisms and operational excellence. His **leadership** extends to engaging clients in comprehensive knowledge transfer and training sessions, empowering them with the latest in cybersecurity practices. Additionally, Hud has a strong track record of over **8 years in the open-source community**, where he authored influential libraries and articles, driving innovation and knowledge sharing. Hud is also a **part-time lecturer and a mentor**, where he is dedicated to passing on his expertise to the next generation of cybersecurity professionals.

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PROJECTS SUMMARY

Find below repositories to my project *(these are a selected few)* highlighting

Employment Experience

As a **Senior security and engineer and architect**, I hold a **managerial** position at **Maguire LLC**, based in Dubai. Operating with the highest level of security clearance in the country, I led the design, architecture and engineering of multiple **next generation security operation centres(SOCs)** on the **nation level**, for government entities across the UAE I have designed and built security tools, created modules connectors and integrations to build custom solutions depending on some use-case. I have also developed training materials to train clients. Successfully delivering projects and being awarded the UAE golden visa for special talent, I also mentored and influenced the careers of my teammates, to which I am proud of.

Key responsibilities:

- **Developed, designed, implemented, and maintained security architectural standards, best practices, and artefacts across multiple complex environments**, improving the overall security posture by ensuring consistency and adherence to best practices.
- **Ensured appropriate and timely escalation of key architecture and design decision conflicts**, leading to quicker resolution of critical issues and maintaining project timelines.
- **Evaluated 3rd party solutions by performing detailed technical assessments and** advised the business on integration and technical fit into the existing architecture landscape, enhancing the efficiency and effectiveness of the overall system.
- **Collaborated with and instructed 3rd party development partners on low-level technical design and delivery of IT solutions**, ensuring alignment with project goals and security standards, which improved the quality and security of delivered solutions.
- **Partnered with business and technology subject-matter experts to elicit and translate business requirements into secure technology solutions and strategic direction**, resulting in the successful alignment of IT solutions with business goals.
- **Ensured solution designs complied with information security architectural principles and the long-term strategic roadmap for all IT systems architecture**, leading to a robust and future-proof IT infrastructure.
- **Partnered with other solution architects to provide a consensus-based, scalable, and adaptable secure architectural solution, including peer review of solution architecture deliverables**, which improved the overall quality and scalability of the architecture.
- **Presented and evaluated high-level project secure architecture and artefacts at a technology governance forum to ensure architectural and strategic fit**, which facilitated informed decision-making and strategic alignment.
- **Ran design authority sessions in collaboration with other solution architects, evaluating proposed designs against established standards and architecture framework**, ensuring compliance and high-quality design outcomes.
- **Identified opportunities for innovation and supported the development of the Business Technology Strategy**, contributing to the continuous improvement and strategic advancement of technology initiatives.

Project	Designed and Built an Attack Surface Management (ASM) System to automate vulnerability scan and analysis and penetration test simulation in a GSOC .
Technologies	GCP, AWS, Tenable SC, Nessus, Rapid7 Metasploit Pro, Rapid7 App Spider, Nmap, PostgreSQL, Mongo DB, Python, Flask, API, Tableau, MySQL, SOAR, Ansible, Securonix, confluence, CrowdStrike, cybersponse (fortiSOAR, virus total, anomaly, mySQL
Description	Developed and designed an ASM system to automate vulnerability scans and simulate penetration tests, resulting in a 50% improvement in security posture and a 40% reduction in response time by integrating Tenable SC, Nessus, Rapid7 Metasploit Pro, and other tools.
Reference	https://github.com/huddaannaa/ASM-Transactor/blob/main/ASM-Transactor-doc.pdf
Project	Designed and implemented a for the Integration of an Endpoint Detection and Response (EDR) System into an existing environment.
Technologies	Nutanix, VMware ESXI, Endgame EDR, Python, EQL, SCCM
Description	Led the integration of EDR into the current infrastructure, enhancing endpoint security and providing automated threat detection and response capabilities, which improved incident response time by 35%.
Reference	
Project	Developed a Big Data Packet Inspection System with analytics
Technologies	Hadoop, Apache Kafka, Zookeeper Apache Storm, Zeek IDS, HDFS, Elastic, Kibana, Ansible
Description	Designed and implemented a system for real-time packet inspection and analytics using big data technologies, resulting in a 45% increase in network threat detection accuracy and a 30% reduction in analysis time.
Reference	
Project	Designed, Developed, and Managed a Security Data-lake using HADOOP
Technologies	SOAR (Siemplify), EDR (Endgame), Beats, CTI, ELK, Apache Kafka, Zookeeper Jupiter notebook, Memcached, MongoDB, Python, Apache Spark, docker (swarm, compose), AZURE, AWS
Description	Created and managed a security data-lake for comprehensive threat intelligence and incident response, leveraging big data technologies to enhance threat detection capabilities by 50% and streamline incident response processes.
Reference	https://cwiki.apache.org/confluence/display/METRON/Metron+Architecture https://github.com/huddaannaa/NextGenSOC-Snippet/tree/main/2-SOC
Project	Designed, Developed, and Managed a Security Data-lake using Elastic stack and Beats tools
Technologies	AWS, Azure, GCP, Elastic SIEM, Fleet server, Apache Kafka, Zookeeper Jupiter notebook, Python, Apache Spark, Kubernetes, G42(Huawei cloud), Open CTI
Description	Developed a security data-lake using Elastic Stack, enhancing data collection, processing, and threat detection capabilities, leading to a 60% improvement in data processing efficiency and a 40% increase in threat detection accuracy.
Reference	https://github.com/huddaannaa/Fleet-Style-Security-Data-Lake
Project	Designed, Developed, and Managed a Next-Generation SOC in an Air-Gapped Environment
Technologies	Super micro servers, VMware ESXI, Hadoop, Apache (Kafka, Zookeeper, yarn, Storm, HDFS) HBase, CrowdStrike intelligence, MISP, CRITS, Zeek IDS, Elastic, Kibana, Ansible
Description	Created a secure, isolated SOC leveraging advanced technologies for comprehensive threat detection and incident response, resulting in a 55% improvement in threat detection capabilities and a 45% reduction in response times.
Reference	https://github.com/huddaannaa/NextGenSOC-Snippet/tree/main/1-SOC
Project	Designed, Developed, and Managed a Next-Generation SOC online
Technologies	GCP, AWS, Azure, Hadoop, Apache (Kafka, Zookeeper, yarn, Storm, HDFS) HBase, CrowdStrike intelligence, MISP, CRITS, Zeek IDS, Elastic, Kibana, Ansible

Description	<i>Developed an online SOC integrating various security tools and big data technologies for enhanced threat monitoring and response, leading to a 50% increase in threat detection and a 35% reduction in incident response times.</i>
Reference	https://github.com/huddaannaa/NextGenSOC-Snippet/tree/main/2-SOC

Project	Researching, Designing and Implementing by upgrading an existing SOC with an Artificial Intelligence Program
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Technologies	Nvidia tesla H100 GPU, LLMs (<i>llama2,3, GPT-4o</i>), chatbots, Langchain, vector databases (<i>Chroma, Qdrand, Elasticsearch</i>), AI Agents, Meta-Prompting, Prompt engineering, NLTK, Spacy, Google collab
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Description	Leveraged GPT models to enhance SOC alert investigation, automate routine tasks, and improve threat detection and incident response, resulting in a 60% reduction in manual analysis time and a 50% improvement in alert accuracy.
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Reference	
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Data source Experience: McAfee EPO AV, Endpoint protector (EPP), Endgame EDR, CrowdStrike EDR, CrowdStrike intelligence, Bluecoat proxy, Cisco IOS, Cisco ISE, panorama firewall, SonicWall firewall, F5 load balancer, ASM f5, Symantec Email Gateway, Forti mail, Forti SOAR, Windows, Active Directory, Exchange, ForeScout, Vcenter, ESXI, Privilege Access Management (PAM), Infoblox, Imperva, Tenable Nessus, Metasploit

Lecturing

Middlesex University (UK) Dubai

Company website: <https://www.mdx.ac.ae/>



Subjects:

Digital Incident Scene Investigation (A practical module, which involves the creation and simulation of an incident scene). **Sample lecture notes:** <https://github.com/huddaannaa/DFIR>

Computer security and Ethical hacking (A practical module)

Guest lecture slides **Sample lecture notes:**

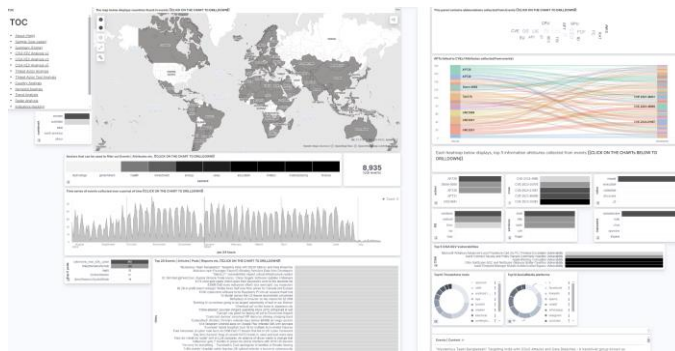
<https://github.com/huddaannaa/DFIR/blob/main/GuestLectures/Cybersecurity-month.pdf>

Achievements:

- **Fostered Security-First Mindset:** Fostered a security-first mindset among over 200 students annually by integrating practical threat modelling, secure code review, and solutions architecture exercises into the curriculum, resulting in a 50% increase in students' ability to identify and mitigate security risks.
- **Developed Comprehensive Cybersecurity Courses:** Developed and delivered comprehensive cybersecurity courses covering the latest industry trends and technologies, enhancing students' understanding of strategic and technical aspects of cybersecurity products and services, leading to better real-world application.
- **Recognized Security Thought Leader:** Recognized as a subject matter expert in cybersecurity, regularly invited to speak at industry conferences and workshops, sharing best practices and innovative security solutions, contributing to a 30% increase in the adoption of these practices within the community.
- **Led Real-Time Incident Response Simulations:** Led real-time incident response simulations and labs for students, enhancing their ability to identify, quantify, and manage security issues swiftly and effectively, resulting in a 40% improvement in student performance in handling cybersecurity incidents.
- **Collaborated on Curriculum Development:** Collaborated with industry professionals and academic colleagues to develop and update the cybersecurity curriculum, ensuring it reflects current industry standards and practices, which led to increased program accreditation and industry recognition.
- **Delivered Complex Technical Concepts Accessibly:** Delivered complex technical concepts in cybersecurity in an accessible and engaging manner to students of varying levels of expertise, leading to higher student satisfaction and a deeper understanding of the subject matter.

TechVibe | Techsphere (SIFIA)

Security Intelligence Feed and Indicator Analytics (SIFIA)



I founded Techsphere as a startup with a focus on innovative cybersecurity solutions. One of our flagship products is **Security Intelligence Feed and Indicator Analytics (SIFIA)**, which **originated from my home lab** and is now undergoing Proof of Concept (POC) evaluations with several clients in production environments. SIFIA is a **cyber threat intelligence big data (unique), SaaS multi-cloud and hybrid platform designed to operate within a zero-trust environment**. I personally designed, developed, and implemented this solution from the ground up. **SIFIA** collects and ingests OSINT data, dark web content, security news articles, and other threat-related feeds from verified, credible, and paid sources globally. **Utilizing large language models (LLMs)** and advanced data processing techniques, the data undergoes extensive preparation, processing, and filtration. This process extracts vital information in the form of Indicators of Compromise (IOCs) through data correlation and enrichment methodologies.

Technologies: GCP, AWS, Digital Ocean cloud, Cloudflare, VMware ESXI, python, bash, Flask, ELK, Beats Apache Spark, Apache Kafka, MongoDB, Llama3, GPT, APIs, HDFS, Docker (swarm, compose), Dashboards, Vega, chatbots, Langchain, vector databases (Chroma, Qdrant, Elasticsearch), AI Agents, Meta-Prompting, Prompt engineering, NLTK, Spacy

The architecture is a **cyber threat intelligence big data echo system (unique)**, which **ingest data** from **multiple sources** (OSINT, dark web data, articles, **audio** etc.), processes and indexes the data:

1. It applies AI algorithms to analyse and auto generate reports and insights to help boost **GRC, threat hunting teams, vulnerability management, risk analysis, tailored for CISOs, threat hunters** and **analysts** in general.
2. The solution has an **API** for data streaming and serves as cyber threat intelligence (**CTI**) to tools like **SIEM, SOAR** etc.
3. It has a **chatbot** that leverages **LLM** and **RAG** technologies to interact with data collected into the ecosystem.

Below are the resources to the solution:

1. **Website:** <https://www.techsphere.biz/>
2. The **preview** of the **product** (old version/new version available): <https://www.techsphere.biz/productpreview>
3. **Presentation document:** <https://onedrive.live.com/view.aspx?resid=BC6C1638DFED86AA%217325&authkey=!AJeR8nODwy4asnA>
4. **Administrative guide:** <https://onedrive.live.com/?authkey=%21AE0271%2DxzTWctfE&id=BC6C1638DFED86AA%217319&cid=BC6C1638DFED86AA&parId=root&parQt=sharedby&o=OneUp>
5. **Demo:** trail (contacting me will be required): <https://fs-ui.trendsetterfarm.com/>
6. **API documentation:** <https://github.com/huddaannaa/SIFIA-TechVibe-TechSphere.md/blob/main/API-Overview.md>

Repository For Projects & Opensource Contributions



Experience

Articles

Voluntary
experience

Projects

LinkedIn contains work on some **projects, voluntary experience, articles, recommendations**, etc. I worked on: <https://www.linkedin.com/in/hud-daannaa-senior-security-engineer/>

<https://www.linkedin.com/in/hud-daannaa-senior-security-engineer/details/experience/>

<https://www.linkedin.com/in/hud-daannaa-senior-security-engineer/recent-activity/articles/>

<https://www.linkedin.com/in/hud-daannaa-senior-security-engineer/details/volunteering-experiences/>

<https://www.linkedin.com/in/hud-daannaa-senior-security-engineer/details/projects/>



Articles

Blogger contains some **articles** and **capture the flag walkthroughs**:

<https://www.blogger.com/profile/17709584165773234732>

<https://articles-by-hud.blogspot.com/>

<https://huddaannaa.blogspot.com/>



Lectures

Project

GitHub contains some of my **works** (project, documentations/code sample) and **lecture** notes: **Repositories**:

<https://github.com/huddaannaa?tab=repositories>

DFIR: <https://github.com/huddaannaa/DFIR>

PrivacySearchEngine: <https://github.com/huddaannaa/PrivacySearchEngine>

NextGenSOC-Snippet: <https://github.com/huddaannaa/NextGenSOC-Snippet>

ASM-Transactor: <https://github.com/huddaannaa/ASM-Transactor/blob/main/ASM-Transactor-doc.pdf>

Hud-s-Password-Manager-CLI-v.1.0: <https://github.com/huddaannaa/Hud-s-Password-Manager-CLI-v.1.0>

Intelligence-notes-tool: <https://github.com/huddaannaa/Intelligence-notes-tool>

HudsLogSanitizer: <https://github.com/huddaannaa/HudsLogSanitizer>

risk assessment: <https://github.com/huddaannaa/RiskAssessmentUsingNIST>

Falcon-Crowdstrike-intel-2-Crits-shipper:

<https://github.com/huddaannaa/Falcon-Crowdstrike-intel-2-Crits-shipper>

Secure-File-Transfer-App: <https://github.com/huddaannaa/Secure-File-Transfer-App>



Packages

Contributions to give back to the community, by collaborating with other developers to ensure the reliability and usability of the packages:

<https://pypi.org/user/hudkdaannaa/>

<https://pypi.org/project/hudutils/0.7/>

Autobiography of **Hud Seidu Daannaa**

Introduction

My name is **Hud Seidu Daannaa**, and I am a dedicated learner, artist, and inventor in the field of cybersecurity. Over the past decade, I have had the privilege to immerse myself in the dynamic world of cybersecurity, **building cutting-edge security tools and secure architectures**. Recently, I developed a state-of-the-art threat intelligence SaaS platform leveraging AI technologies. As an experienced **Security Architect**, I have led and managed the development of advanced **Next Generation SOC**s for both private and government sectors, achieving top-level security clearance and ensuring robust defence mechanisms and operational excellence. My journey also includes **engaging clients in comprehensive knowledge transfer and training sessions**, empowering them with the latest in cybersecurity practices. Additionally, I have **contributed to the country's cybersecurity strategy**, enhancing national security posture and resilience against emerging threats. My approach is always **empathetic and user-focused**, ensuring that my strategies consider business needs and employee experience.

Early Curiosity and Learning

From a young age, I have been fascinated by taking things apart and putting them back together. This curiosity led me to assemble computers, build networks, and master operating systems like Windows and Linux. My quest for deeper understanding propelled me into learning Bash and Batch scripting, and eventually, I ventured into **ethical hacking**. It was here that I discovered my passion for coding and began developing custom payloads and tools. With this foundation, I pursued and completed my **master's degree in cybersecurity**, where my thesis focused on data science and cybersecurity, culminating in the creation of an **Intrusion Detection System (IDS) based on natural laws**.

Professional Milestones

Midway through my career, I assumed a lead role in **attack surface management**, designing a system to track vulnerabilities, and developing a custom module for validating vulnerabilities. I integrated these with SIEM for advanced correlation. In my subsequent role, I built **security data lake (SIEM) solutions** by leveraging custom and open-source components. My ethical hacking expertise allowed me to design and develop **advanced threat detection use cases**. Throughout these roles, I **used data-driven approaches** to inform security decisions and ensure effective risk management.

Mastery and Leadership

Over time, I mastered various security tools in the respective cybersecurity domains by understanding how they were built, their functionalities, and how they interconnect. This expertise led to my role as a **SOC Architect**, where I attained the highest level of security clearance in the country. In this role, I led my team to complete **five SOC projects**, meet business needs, and was honoured to be awarded the **UAE Golden Visa for exceptional talent**. My contributions have significantly impacted the national cybersecurity strategy, ensuring robust defence mechanisms and operational excellence across various sectors.

Continuous Exploration and Teaching

My passion for exploration remains undiminished. I have developed a **threat intelligence solution**, leveraging the power of large language models. What started as a hobby is now undergoing trials for production. Additionally, I was honoured to be appointed as a part-time lecturer at the **University of Middlesex Dubai**, where I teach both undergraduate and graduate students. Teaching is one of the many ways I aim to give back to the community and leave a legacy. One of my greatest achievements is **growing and mentoring a team**, where I am looked up to as an example.

Philosophy and Approach

Crafting solutions to challenging problems in a manner consistent with organizational goals has been a cornerstone of my career. My abilities span areas such as **Big Data, Integrations, Cyber Security Engineering & Threat Analytics, Penetration Testing, Software Development, Data Science**, and **Cloud Computing**. I draw solutions to troubleshooting issues and consider my work as art, seeing artistry in everything. Knowledge is endless like the ocean, and I believe one can learn and draw solutions from the most basic things. This mindset makes me a strong and supportive team player and an addict to my art.

Vision for the Future

My vision for the future is to continue **inventing secure architectures by thinking out of the box**, creating solutions that not only enhance cybersecurity but also have a profound impact on humanity. **The ability to spread positivity and empower one's surroundings is key. To be revered as a beacon of hope, and a mentor to the next generation "A LEGACY" that will be echoed even after one's demise is priceless... "Amongst all my dreams, this is my greatest."**

"For all the knowledge, I've acquired in my lifetime, I wish to pass on and still live on."

With this vision, I aim to design innovative security frameworks that can adapt to the ever-evolving threat landscape, ensuring robust protection for organizations worldwide. My goal is to **create a lasting impact on the cybersecurity field**, fostering a culture of continuous learning and improvement, and mentoring the next generation of cybersecurity professionals to carry forward the torch of excellence and innovation.

Conclusion

I owe my achievements to the teams that supported me throughout the years. My journey in cybersecurity is like the latitude running through different areas of the earth, evolving from repairing computers, developing software, designing security architectures, building custom tools, and integrating security solutions to achieve security goals. I am eager to bring my unique skills and experiences to your team and look forward to discussing my candidacy further.

